



Machine learning for water quality prediction and uncertainty assessment

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ABSTRACT

Effective water quality monitoring requires predictive models that combine high accuracy, interpretability, and credible uncertainty quantification. Machine learning (ML) techniques have emerged as powerful tools for predicting water quality and quantifying associated uncertainties. Similarly, Bayesian deep learning (BDL) frameworks enable probabilistic predictions that quantify uncertainties. These approaches can capture nonlinear interactions and provide robust predictions in diverse environmental conditions. This study integrated ensemble ML and BDL to assess the complex relationships between physicochemical parameters and the Water Quality Index (WQI). Six supervised ensemble ML algorithms, namely Decision Tree (DT), Random Forest (RF), Extra Trees (ERT), XGBoost, CatBoost, and LightGBM were evaluated using Bayesian optimisation to identify the optimal hyperparameter configurations. DT achieved the highest predictive accuracy with MAE = 0.657 and 0.428, RMSE = 1.181 and 0.747, MAPE = 10.561 and 7.155, $R^2 = 0.960$ and 0.987 , and nRMSE = 0.065 and 0.042 for the training and test sets, respectively. The DT outperformed more complex ensemble models, and SHapley Additive exPlanations (SHAP)-based eXplainable Artificial Intelligence (XAI) identified the most influential predictors, aligning model predictions with underlying hydrochemical processes. To capture predictive uncertainty, a probabilistic BDL was developed yielding probabilistic outputs and explicit epistemic uncertainty estimates. ROC analysis confirmed strong performance across WQI classes, with AUC scores of up to 0.90 for WQI classes. The probabilistic approach provides actionable insights for adaptive water quality management, enabling targeted monitoring in areas of high uncertainty and supporting transparent, evidence-based decision-making. These results underscore the value of integrating ML, and Bayesian optimisation to advance robust and adaptive water quality assessment. The proposed workflow provides a scalable framework to enhance monitoring, optimize resources, and advance sustainable water management aligned with the SDGs.

1. Introduction

Water quality degradation poses an escalating threat to environmental sustainability, economic development, and human health (du Plessis, 2022; Scanlon et al., 2017). Globally, the lack of access to safely managed drinking-water systems is linked to a wide spectrum of adverse health and ecological impacts (WHO, 2021). In vulnerable regions, uncontrolled human activities have exacerbated contamination risks and stressed institutional capacity for monitoring and adaptive management (Zhu et al., 2022; Qadir et al., 2006). Both conventional and emerging contaminants increasingly threaten water resources, posing serious risks to ecosystems and human health (Shaji et al., 2024). Conventional pollutants persist in aquatic environments and are major drivers of water quality degradation, strongly linked to ecological deterioration, eutrophication, and adverse health outcomes (Assegede

et al., 2022; Dudgeon et al., 2006; Morita et al., 2021). Emerging contaminants, frequently bioaccumulative and insufficiently regulated, present escalating risks to environmental and public health (Egbueri et al., 2024; Li et al., 2024; Mora et al., 2021). Those pollutants increasingly contribute to water quality degradation; however, routine operational monitoring and regulatory frameworks continue to rely predominantly on physicochemical indicators aggregated through Water Quality Index (WQI) based approaches (Brown et al., 1972). Thus, predictive modeling of WQI remains essential for translating complex and evolving contaminant pressures into actionable, management relevant assessments, particularly in regions facing monitoring gaps, data scarcity, and environmental uncertainty (Chidiac et al., 2023; Misstear et al., 2023; Uddin et al., 2021).

Changes in hydro-biogeochemical processes and stream-aquifer interactions undermine established theories and models of riverine system

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dynamics, intensifying pollutant movement and complicating water quality management (Fabian et al., 2023; McDonald et al., 2011). The WQI integrates diverse physicochemical indicators to assess and interconnect water quality (Brown et al., 1972). Advancing WQI prediction enhances monitoring and management capabilities, as various contaminants are pervasive and consequential to environmental health, contaminating aquifer systems (Ascott et al., 2017). Though, index-based methods are prevalent, typically rely on static parameters, limited observations, and subjective weighting schemes, which can introduce biases and underrepresent inherent uncertainty (Uddin et al., 2021, 2023a). Given their high movement in unsaturated zones and long residence times in aquifer systems, contaminants present challenges for mitigation and long-term monitoring (Burow et al., 2010). Therefore, accurate predictions and assessments are essential for sustainable water management and effective public health protection (Deng et al., 2023). Conventional approaches usually rely on empirical models and geo-statistical interpolations to estimate contamination patterns (Rivett et al., 2008), constrained by their stationarity and limited flexibility in capturing nonlinear interactions among hydrogeological, climatic, and anthropogenic factors (Beven, 2016).

Recently Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL) techniques has proven effective for modeling water quality by extracting patterns from historical data and validating prediction performance (Ahmed et al., 2024; Zhi et al., 2024). In parallel, advanced AI techniques are increasingly used to assess and quantify model uncertainty, thereby supporting more reliable and transparent decision-making (Abdar et al., 2021; Uddin et al., 2021). Despite this progress, current ML models for WQI prediction face challenges such as hyperparameter tuning, adaptation to regional variability, and maintenance of flexibility (Zheng et al., 2023). Although ensemble learning shows promise for addressing these issues (Wang et al., 2022; Zhu et al., 2022), its application in water quality modeling is limited. In response, ML techniques have been emerged as favourable, offering the capacity to integrate heterogeneous datasets, model complex dependencies, and deliver high-resolution predictions with minimal parametric assumptions (Uddin et al., 2021, 2022a). ML techniques have shown strong performance in environmental contamination studies, often outperforming predictive accuracy and generalizability (Adediji et al., 2022; Al-Sulttani et al., 2021; Uddin et al., 2022a, 2023b). These consist of tree-based ensemble methods, Decision Tree (DT), Random Forest (RF), Extremely Randomised Trees (ERT) (Biau and Scornet, 2016; Geurts et al., 2006; Quinlan, 1986), and other advanced gradient boosting frameworks: Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Categorical Boosting (CatBoost) used in environmental modeling, hydro-geochemical investigations, and large-scale groundwater monitoring datasets (Chen and Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018). Recent studies have demonstrated the effectiveness of tree-based ensemble methods for mapping and risk classification (Chen and Guestrin, 2016; Prokhorenkova et al., 2018), forecasting nitrate leaching under different land-use scenarios, and identifying critical source areas (Huang et al., 2021). However, despite these advances, challenges persist in hyperparameter optimisation, interpretability, and incorporation of spatiotemporal uncertainty into predictive workflows.

Many existing predictive approaches are unsuitable for operational decision-making in data-scarce or rapidly changing environments (Patterson et al., 2022; Uddin et al., 2022b). Addressing these limitations requires a paradigm shift towards probabilistic and data-driven frameworks that leverage current developments in ML and Bayesian inference to deliver more reliable, transparent, and uncertainty-aware water quality assessments (Ghahramani, 2015; Xu et al., 2021). Bayesian Deep Learning (BDL) has emerged as a powerful methodological innovation that not only enhances the predictive performance in high-dimensional, nonlinear systems, but also quantifies both aleatoric and epistemic uncertainties (Gal and Ghahramani, 2016; Hüllermeier and Waegeman, 2021; Kendall and Gal, 2017). No prior study has

systematically quantified how each modeling stage contributes to total uncertainty, leaving significant gaps in understanding how uncertainty propagates across diverse water bodies and under varying data conditions (Abbasi and Abbasi, 2012). Many studies have evaluated various WQI models and examined how their structural differences influence outcomes, underscoring the need to quantify uncertainties throughout the modeling process to enhance reliability (Abbasi and Abbasi, 2012; Uddin et al., 2021; Zhu et al., 2022). However, the applications of BDL are still emerging, and initial studies have shown its potential to outperform traditional neural networks while providing probabilistic insights that are critical for risk-informed decision making (Han et al., 2025).

Despite significant advances in ML-based WQI prediction, most existing studies remain largely deterministic, focusing primarily on predictive accuracy while offering limited treatment of uncertainty and interpretability. Previous ML applications have demonstrated strong performance in estimating WQI from physicochemical parameters; however, they rarely quantify epistemic uncertainty, provide limited insight into feature relevance, and seldom integrate uncertainty into decision relevant frameworks. Recent applications have demonstrated the efficacy of Bayesian neural networks in environmental prediction tasks (Blundell et al., 2015), where propagating uncertainty improves reliability and supports decision-making (Talukdar et al., 2023). BDL improves prediction accuracy and explicitly quantifies uncertainties in model design and input selection (Gal and Ghahramani, 2016). However, no prior study has systematically integrated ML models, model interpretability, and BDL into an integrated workflow for WQI prediction with explicit uncertainty quantification limiting both predictive reliability and transparency in water quality assessments. Moreover, although both conventional and emerging contaminants contribute to water quality degradation, operational monitoring and WQI-based assessment frameworks in many regions still rely predominantly on conventional physicochemical indicators (Brown et al., 1972). Consequently, predictive modeling of WQI remains essential for translating complex and evolving contaminant pressures into actionable, management relevant assessments, particularly in regions facing monitoring gaps, data scarcity, and growing environmental uncertainty (Chidiac et al., 2023; Misstear et al., 2023; Uddin et al., 2021).

The main objectives of this study are to: (1) evaluate the predictive performance of machine learning models for WQI estimation using Bayesian optimisation, (2) identify key physicochemical drivers through SHAP-based interpretability, and (3) develop a BDL framework to quantify uncertainty that can support adaptive and uncertainty-aware water quality management. These insights support risk-aware and adaptive monitoring in complex wastewater environments, such as aquaculture systems. This study presents a unified, data-driven framework that integrates BDL, ensemble machine learning, and SHAP-based interpretability to probabilistically predict WQI while explicitly quantifying uncertainty supporting adaptive water quality management. For practitioners and regulators, this uncertainty-aware approach enables risk-based resource allocation, and prioritizing monitoring targets. The framework strengthens the robustness and credibility of water quality predictions, supports the prioritisation of monitoring and remediation, informs policy decisions under environmental and socio-economic change, and advances adaptive water governance in pursuit of Sustainable Development Goal 6 (Clean Water and Sanitation).

2. Materials and methods

2.1. Data description and featuring

2.1.1. Datasets and composite indices

The dataset employed in this study comprised comprehensive physicochemical measurements of groundwater samples systematically collected from regional monitoring networks spanning diverse hydrogeological settings in the Gilgel Gibe Catchment (Mengistu et al., 2025),

Southwestern Ethiopia. It encompasses both upland and lowland aquifer zones (Mengistu et al., 2022). Each observation was georeferenced with precise latitude and longitude coordinates, enabling the spatial modeling of contamination patterns across the study area. The input variables included a suite of parameters widely recognised in water quality assessment such as pH, electrical conductivity (EC), total dissolved solids (TDS), and the concentrations of major cations and anions, specifically sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), magnesium (Mg^{2+}), chloride (Cl^-), sulfate (SO_4^{2-}), bicarbonate (HCO_3^-), nitrate (NO_3^-), and fluoride (F^-). The derived ionic ratios, including SAR, sodium-to-chloride ratio (Na^+/Cl^- ratio), calcium-to-magnesium ratio ($\text{Ca}^{2+}/\text{Mg}^{2+}$ ratio), and chloride-to-bicarbonate ratio ($\text{Cl}^-/\text{HCO}_3^-$ ratio), were incorporated to capture the geochemical processes known to influence nitrate movement.

Composite indices, including the percentage of sodium and total water hardness, were also included to enrich the feature space with integrative measures of water rock interactions and anthropogenic impacts based on the thresholds of the WHO guidelines (WHO, 2017). Composite ionic indices were calculated from major cation and anion concentrations to provide integrative indicators of hydrochemical processes. All indices were derived using ion concentrations expressed in milliequivalents per liter (meq L^{-1}). The Sodium Adsorption Ratio (SAR) quantifies the sodicity hazard of water by evaluating the dominance of sodium relative to calcium and magnesium. Higher SAR values indicate greater potential for cation-exchange reactions, clay dispersion, and reduced hydraulic conductivity, all of which affect nitrate transport; SAR is computed as (Eq. (1)). Percentage Sodium ($\text{Na}\%$) provides a complementary measure of relative sodium enrichment and is widely used for evaluating irrigation water quality; it incorporates major cations into a single expression as (Eq. (2)). Total Hardness (TH), expressed as CaCO_3 (mg L^{-1}), represents the combined contributions of calcium and magnesium and is important for interpreting carbonate equilibria and buffering capacity. TH is calculated using standard molecular-weight and charge-based conversion factors as shown in (Eq. (3)).

$$\text{SAR} = \frac{[\text{Na}^+]}{\sqrt{\frac{[\text{Ca}^{2+}] + [\text{Mg}^{2+}]}{2}}} \quad (1)$$

$$\text{Na}(\%) = \frac{[\text{Na}^+] + [\text{K}^+]}{[\text{Ca}^{2+}] + [\text{Mg}^{2+}] + [\text{Na}^+] + [\text{K}^+]} \times 100 \quad (2)$$

$$\text{TH} (\text{mg/L}) = 2.497 \times [\text{Ca}^{2+}] + 4.115 \times [\text{Mg}^{2+}] \quad (3)$$

These coefficients convert Ca and Mg from mg/L to their hardness equivalents, providing a universally comparable hardness metric. To further characterise geochemical processes and analytical ion ratios were derived: $\frac{\text{Na}^+}{\text{Cl}^-}, \frac{\text{Ca}^{2+}}{\text{Mg}^{2+}}, \frac{\text{Cl}^-}{\text{HCO}_3^-}$. The Na^+/Cl^- ratio helps distinguish between halite dissolution, ion-exchange, and anthropogenic sodium inputs; $\text{Ca}^{2+}/\text{Mg}^{2+}$ ratio informs carbonate–dolomite dissolution pathways and mixing of water sources; and The $\text{Cl}^-/\text{HCO}_3^-$ ratio is a sensitive indicator of salinization processes and human impacts such as sewage or fertilizer intrusion. These indices and ratios provide a multidimensional representation of hydrochemical dynamics, enhancing the predictive value of the feature space and enabling more robust interpretation of nitrate behavior within the modeled system. All thresholds for explanatory guidance follow the WHO (2017) water quality guidelines. The water quality index (WQI) is a well-recognised metric for evaluating water quality and guiding management decisions (Abbasi and Abbasi, 2012; Han et al., 2025). To empirically compute an overall weighted WQI, each water quality parameter was assigned a relative weight (Eq. (4)), and each sample was rated on a scale (Eq. (5)) providing a single composite index value (Eq. (6)).

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (4)$$

$$q_i = \frac{C_i}{S_i} \times 100 \quad (5)$$

$$\text{WQI} = \sum_{i=1}^N W_i q_i \quad (6)$$

Where W_i is the relative weight, w_i is the weight of each parameter, n represents several parameters, q_i is the quality rating, C_i is the concentration of each pollutant in the physiochemical parameter in every water sample in mg/L, and S_i WHO standard concentration.

2.1.2. Data preprocessing

Prior to model development, the dataset was subjected to a standardized preprocessing procedure to ensure consistency and robustness. Missing values were examined across all physicochemical variables; observations with incomplete records were excluded due to their limited proportion relative to the total dataset. The dataset exhibits substantial statistical heterogeneity and reflects the intrinsic complexity of groundwater hydrogeochemical systems. The physicochemical variables span wide numerical ranges and demonstrate markedly different distributional behaviors, indicative of nonlinear geochemical controls and spatially heterogeneous processes. Certain parameters, such as pH, remain constrained within a relatively narrow and buffered range due to carbonate equilibrium and natural regulatory mechanisms. In contrast, most dissolved constituents reflect geochemical enrichment processes, including mineral dissolution, ion exchange, evaporative concentration, and localized anthropogenic influence. The dataset is characterized by strong scale disparities among variables, with some parameters varying over orders of magnitude relative to others. Such heterogeneity reflects the coexistence of low-concentration trace constituents alongside high-magnitude integrative indicators such as EC, TDS, and bicarbonate. This diversity in scale and dispersion necessitates normalization to ensure numerical stability and balanced feature representation during model training.

Several variables exhibit distributional features indicative of hydrogeochemical regime mixing, likely arising from spatial variability, differences in aquifer lithology, and varying groundwater residence times. This implies that the dataset cannot be adequately represented using single-mode or linear statistical assumptions. Instead, it embodies multiple coexisting water quality states that require flexible, nonlinear modeling approaches capable of capturing regime dependent relationships. Notably, the dataset contains extreme but hydrochemically plausible observations that reflect natural system variability rather than noise or measurement error. Retaining these values preserves critical information about boundary conditions and stress states within the aquifer system, which are essential for reliable prediction and uncertainty characterization. Accordingly, robust learning algorithms and probabilistic frameworks were adopted to explicitly account for uncertainty propagation. All continuous input features were standardized to zero mean and unit variance to facilitate model convergence and ensure comparability across predictors. Outliers were assessed using inter-quartile range analysis and domain-informed thresholds. However, no extreme values were removed, as the observed ranges were consistent with known hydrogeochemical processes. No temporal filtering was applied due to the cross-sectional nature of the dataset. Spatial information was incorporated exclusively through positional encoding to avoid direct spatial leakage while preserving spatial variability. Overall, the dataset represents a challenging environmental system characterized by scale heterogeneity, multimodal behavior, and meaningful extremes. These properties justify the adoption of feature standardization, robust ensemble machine learning (ML) methods, explainable modeling

strategies, and uncertainty quantification to ensure both predictive accuracy and technically logical inferences.

2.1.3. Multi-scale spatial feature engineering

Accurate prediction of water quality across heterogeneous landscapes requires ML models to capture both spatial variability and site-specific heterogeneity; hence, all numeric features were scaled to standard distributions, aiding convergence and ensuring that feature contributions are comparable (de Amorim et al., 2023; Pinheiro et al., 2025; Suwadi et al., 2022). In addition to physicochemical parameters, spatial coordinates were transformed using positional encodings to capture spatial gradients and periodicities in the contaminant distribution, which raw coordinates alone could not adequately represent. The use of sinusoidal spatial encoding enhances spatial representation without relying on traditional geostatistical methods. These transformations enabled the modeling framework to account for unobserved spatial processes such as proximity to recharge areas, agricultural hotspots, and point sources of contamination (Suwadi et al., 2022). Classical ML features such as raw latitude, longitude, and water quality parameters are not inherently well suited for learning periodic, multi-scale spatial dependencies (de Amorim et al., 2023; Pinheiro et al., 2025). Directly using geographic coordinates as numerical inputs often results in poor generalization, since models may misinterpret absolute values rather than relative positions.

To overcome this limitation, this study transformed spatial coordinates using Sinusoidal Positional Encoding (SPE) and Multi-Scale Spatial Positional Encoding (MSPE) strategies. These encodings project spatial and water quality inputs into a bounded, multi-scale feature space. SPE and MSPE enhance spatial representation without relying on traditional geostatistical methods, preserving both coarse- and fine-scale variability along non-linear trends and enabling the predictive network to access spatial information across multiple resolutions. This approach builds on advances in neural network positional encoding (Tancik et al., 2020), Fourier-based feature mappings for spatial representation (Li et al., 2021), and recent applications in environmental and geospatial tasks (Karpatne et al., 2017). Latitude, longitude, site identifiers, and transformed water quality features are each mapped into a high-dimensional sinusoidal space, where every scalar is expanded into a vector of sine and cosine terms at geometrically scaled frequencies. For spatial coordinates, latitude and longitude are encoded separately and concatenated. Site identifiers are embedded using positional encoding, yielding a continuous and smooth feature space that avoids sparsity and enhances generalization. Each water quality variable is likewise encoded using sinusoidal mapping to preserve nonlinear thresholds and relationships in water chemistry. For spatial grids of two-dimensional coordinates, SPE uses sine and cosine functions at exponentially increasing wavelengths to capture both short- and long-range positional patterns. MSPE augments SPE by enabling hierarchical spatial representation, improving generalization to variable input resolutions and enhancing robustness in hydro-climate and environmental modeling, where processes are inherently multi-scale. These transformations allow the framework to capture complex spatial water quality dynamics and account for unobserved processes such as proximity to recharge areas, agricultural hotspots, and point sources of contamination. This capability provides a reliable mechanism for producing water-quality predictions, facilitating the prioritisation of monitoring efforts, and supporting evidence-based planning and management. For a spatial location (x, y) , the SPE is defined as Eqs. (7 & 8) and the MSPE extends this by concatenating SPEs computed at different scales $s_1, s_2, \dots, s_k$ as define in (Eq. (9)).

$$PE_{\{x,2i\}} = \sin\left(\frac{x}{10000\left(\frac{2i}{d}\right)}\right), PE_{\{x,2i+1\}} = \cos\left(\frac{x}{10000\left(\frac{2i}{d}\right)}\right) \quad (7)$$

$$PE_{\{y,2i\}} = \sin\left(\frac{y}{10000\left(\frac{2i}{d}\right)}\right), PE_{\{y,2i+1\}} = \cos\left(\frac{y}{10000\left(\frac{2i}{d}\right)}\right) \quad (8)$$

$$MSPE_{x,y} = \sum_{k=1}^K PE^{(s_k)}(x, y) \quad (9)$$

Where d = embedding dimension; i, j = channel indices for x and y ; each PE^{s_k} is a sinusoidal encoding at resolution s_k .

2.2. Model development and evaluation

2.2.1. Machine learning model training

At this stage, supervised machine learning models were developed to estimate prediction applications from predictor variables based on their demonstrated success in modeling structured environmental datasets (Adedeji et al., 2022). Ensemble and boosting methods, in particular, have been shown to have robust performance in environmental studies, outperforming predictive accuracy and generalizability (Adedeji et al., 2022; Al-Sulttani et al., 2021; Uddin et al., 2022a, 2023b). In this study, six ML models provided a robust technical foundation for capturing nonlinear dependencies and complex interactions among the physicochemical variables governing water quality processes. These comprise of DT, RF, and ERT (Biau and Scornet, 2016; Geurts et al., 2006; Quinlan, 1986); XGBoost, LightGBM, and CatBoost (Chen and Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018) used in environmental modeling, hydrogeochemical investigations, and large-scale monitoring datasets. The ML workflow (Fig. 1) framework for water quality prediction and architecture of probabilistic WQI using physicochemical, cation, anion, and geochemical ratio datasets. The process begins with data preprocessing and feature selection to identify the most relevant input parameters for the model. Six supervised ML algorithms, including RF, DT, ERT, XGBoost, CatBoost, and LightGBM, were trained and optimised using Bayesian hyperparameter tuning to maximise predictive performance. The dataset was split into training (70 %), validation (15 %), and test (15 %) subsets using stratified sampling to preserve the distribution of WQI classes. Hyperparameter tuning was conducted on the training and validation sets, and results were reported on the independent test set. Following model training, a performance evaluation phase assesses the models using standard metrics to enable optimal model selection.

2.2.2. EXplainable Artificial Intelligence (XAI)

SHAP (SHapley Additive exPlanation) analysis provided evidence of consistent learning and meaningful feature prioritisation. SHAP used to enhance transparency and quantify the contribution of each predictor to the model outputs (Lundberg and Lee, 2017), providing insights into whether the model reasoning was aligned with the established hydrogeochemical understanding. To enhance interpretability, the framework integrates XAI through SHAP, quantifying each feature's contribution to the predictions. Feature selection combined domain knowledge (hydrogeochemical relevance) with SHAP values threshold iterative feedback. This introduces a feedback loop from the SHAP analysis back to feature selection, enabling iterative refinement of input features (Fig. 1). As illustrated in Fig. 1, the process began by training an initial model using all candidate predictors, including spatial coordinates transformed using SPE and MSPE to prevent any spatial leakage during model setup. These encodings project raw latitude-longitude information into a bounded multi-frequency representation that avoids direct geometric memorization.

SHAP values were computed for each feature, ranked by mean absolute contribution, and features with consistently negligible influence defined as those falling in the lowest decile of SHAP importance across successive refinement cycles were removed. The model was retrained

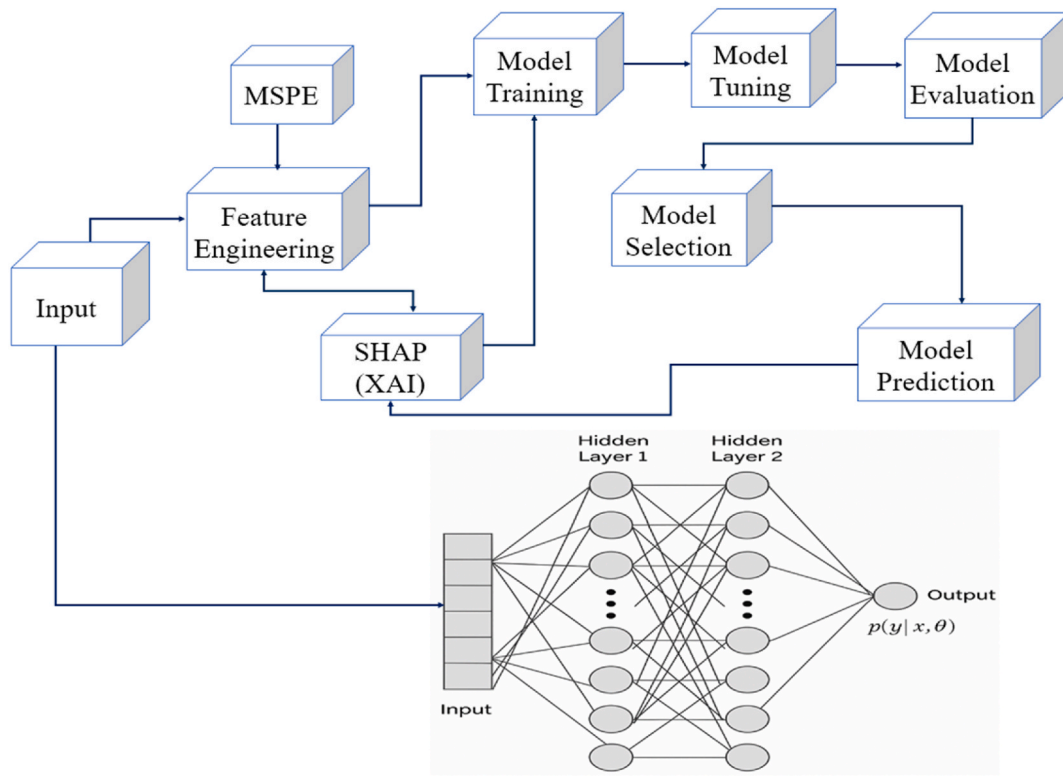


Fig. 1. Explainable machine learning workflow for water quality prediction with SHAP-guided feature refinement and conceptual architecture for probabilistic learning.

after each iteration, and the refinement continued until the SHAP ranking stabilized between consecutive iterations, which served as the stopping criterion. The final features comprised of the dominant water quality variables and SPE/MSPE-derived spatial encodings. This closed-loop design ensures robust and reliable predictions, making the system valuable for environmental monitoring and decision-making. Model explainability establishes a robust framework for developing predictive models that can support informed decision-making in water quality assessment. The architecture supports data-driven, interpretable water quality assessments while promoting adaptive learning and continuous improvement in model reliability, which is critical for achieving SD6 (clean water and sanitation) and advancing smart water resource management under conditions of uncertainty.

2.2.3. Bayesian optimisation

The predictive performance of the model depends on hyperparameter settings that balance bias, variance, and computational efficiency (Probst et al., 2019). Hyperparameter tuning reduces under-fitting (high bias) and over-fitting (high variance) by adjusting model complexity and flexibility, thus governing the error balance in supervised learning via the bias–variance trade-off (Guan and Burton, 2022). Hyperparameter optimisation can thus reduce generalisation error, but overly complex searches suffer further computational cost, requiring accuracy in cost-benefit trade-offs (Bischl et al., 2023; Guan and Burton, 2022). This study employed Bayesian optimisation hyperparameter tuning to ensure the robustness and reproducibility of the predictive models. Bayesian optimisation used to tune the critical hyperparameters, and further enhancing convergence and performance (Oliveira et al., 2022; Snoek et al., 2012). Bayesian optimisation constructs a probabilistic surrogate model, typically a Gaussian Process of the objective function that maps hyperparameters to prediction errors, and iteratively selects value configurations expected to improve performance (Snoek et al., 2012). The optimisation process began with randomly sampled arrangements to initialize the surrogate model,

followed by iterations guided by the expected improvement acquisition function. Bayesian optimisation places a prior distribution over the objective function f and sequentially updates it with observed data, thereby refining the search toward more promising regions of the hyperparameter space (Snoek et al., 2012). An acquisition function $\alpha(x|D_t)$ is defined to guide the selection of the next query point at each iteration as a sequential decision rule using (Eq. (10)). The acquisition function α balances exploration (learning about uncertain regions of f) and exploitation (refining the estimate near promising optima).

$$X_{t+1} = \arg \max_{x \in X} \alpha(x|D_t) \quad (10)$$

After each iteration, the models were retrained and validated, and the model performance was evaluated using multiple complementary metrics. Of these, the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) quantified the average predictive error (Chai and Draxler, 2014; Hodson, 2022; Willmott and Matsuura, 2005), whereas the Mean Absolute Percentage Error (MAPE) indicated the error relative to the observed variables (Goodwin and Lawton, 1999; Hyndman and Koehler, 2006; Willmott and Matsuura, 2005). The R-squared (R^2) measures the proportion of variance explained (Chicco et al., 2021), and the normalised RMSE (nRMSE) enables scale-independent comparisons (Kedam et al., 2024). The mathematical can be compute using given (Eq. (11)–(15)) as RMSE (Eq. (11)), MAE (Eq. (12)), MAPE (Eq. (13)), R^2 (Eq. (14)), and nRMSE (Eq. (15)). All metrics were computed on an independent test set to provide unbiased estimates of generalization performance. To diagnose overfitting, the performances of the training and test sets were compared, and models showing large discrepancies were subjected to further examination.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (11)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (12)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (13)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (14)$$

$$nRMSE_{range}(\%) = 100 \times \frac{RMSE}{\max(y_i) - \min(y_i)} \quad (15)$$

where y_i represents the observed value of the variable at observation i , and $\hat{y}_i$ denotes the corresponding predicted value. The term $\bar{y}$ is the mean of the observed values, and n is the total number of observations used in the analysis. The symbols $\max(y_i)$ and $\min(y_i)$ indicate the maximum and minimum values of the observed data, respectively. These statistical indicators quantify model accuracy, bias, and explanatory performance across varying data magnitudes and scales. All metrics were calculated on an independent test set to provide unbiased estimates of model generalization. Overfitting was assessed by comparing training and test errors; models exhibiting substantial discrepancies were examined further until the optimal parameter configurations were identified as indicated in (Table 1).

2.3. Probabilistic machine learning

To model the complex, nonlinear relationships between physicochemical predictors and WQI categories while explicitly quantifying predictive uncertainty. Then, the study developed a BDL, which uses probabilistic layers that capture uncertainty over weights and activations and is trained using Bayesian inference (Jospin et al., 2022). This probabilistic framework enables the estimation of posterior distributions over model parameters rather than point estimates (Saranathan et al., 2024), supporting uncertainty-aware predictions that are essential for risk-informed water-quality management. The BDL model was trained using variational inference, wherein the posterior distribution over network weights was approximated by a tractable variational distribution. The BDL architecture consists of an input layer corresponding to standardized physicochemical variables. The prior distributions were

specified as isotropic zero mean and unit variance, which is consistent with the established Bayesian practices (Blundell et al., 2015). The target variable was the WQI class, which was categorised into five ordinal levels: Excellent, Very Good, Good, Poor, and Very Poor, following the established classification criteria (Brown et al., 1972). Labels were encoded numerically as Class (0–4) to support ordinal classification modeling, where the DenseVariational output layer with Softmax activation yielded predictive class probabilities, enabling a probabilistic interpretation of the WQI (Fig. 1). Model optimisation was performed by maximizing the Evidence Lower Bound (ELBO), defined as the expected log-likelihood of the observed labels minus the Kullback–Leibler (KL) divergence between the variational posterior and the prior distributions. To obtain probabilistic predictions from the BDL model, Monte Carlo (MC) sampling was employed during inference.

MC sampling was used to generate probabilistic WQI predictions by repeatedly sampling input parameters from their respective probability distributions across numerous iterations. MC sampling is a stochastic uncertainty propagation technique that quantifies how input variability influences model outputs (Beven, 2016; Refsgaard et al., 2007). Because the variational layers learn probability distributions over the network weights rather than single point estimates, each forward pass samples a different understanding of these weights from the learned posterior. By repeating this stochastic forward propagation multiple times, an empirical distribution of the predicted WQI classes is generated, from which mean predictions, variances, and confidence intervals are derived. This MC procedure is a standard component of variational Bayesian inference and requires no specific computational resources it simply executes repeated forward passes through the trained model to approximate the posterior predictive distribution. Each random draw produced a distinct realization of the WQI, and the resulting ensemble of simulations was used to compute mean values, variance, and confidence intervals. This approach provides a robust estimate of predictive uncertainty, allowing the model to represent natural variability and measurement noise while ensuring reproducibility and transparency in uncertainty quantification.

2.4. Variational inference framework

The Bayesian neural network architecture consists of an input layer comprising standardized physicochemical variables and engineered spatial features, followed by multiple fully connected variational layers. The given framework (Fig. 2) provides a reliable mechanism for producing uncertainty-aware water quality predictions, facilitating the prioritisation of monitoring efforts and supporting evidence-based policy development. The output layer employs a Softmax activation function to produce class probability distributions for the ordinal WQI categories. Unlike conventional neural networks, all weights and biases in the model are treated probabilistically rather than as fixed point estimates. This formulation enables the explicit representation of epistemic uncertainty arising from limited data availability and model structural uncertainty. To facilitate probabilistic prediction and uncertainty quantification, the BDL model was formulated within a variational inference framework. The training dataset be denoted by $D = \{(x_i, y_i)\}_{i=1}^N$, where x_i represents the vector of standardized physicochemical predictors and engineered spatial features, and y_i denotes the corresponding WQI class label. In the Bayesian neural network formulation, the model parameters θ , comprising all weights and biases, are treated as random variables. Bayesian inference is then used to estimate the posterior distribution $p(\theta|D)$, which quantifies uncertainty in the model parameters conditioned on the observed data.

Variational inference was achieved maximizing ELBO, which was computed as the sum of the expected log-likelihood of the observed labels and a complex penalty defined by the KL divergence between the variational posterior and prior (Kristiadi et al., 2019). This formulation ensures a principled trade-off between the model fit and complexity,

Table 1
Hyperparameter search spaces and optimal model configurations summary.

Model	Hyperparameter	Ranges	Best Parameters
RF	n_estimators	10–200	112.08
	max_depth	2–20	5
	min_samples_split	2–20	10
	min_samples_leaf	1–20	3
DT	max_depth	2–20	7.44
	min_samples_split	2–20	4.64
	min_samples_leaf	1–20	2.75
ERT	n_estimators	10–200	23.93
	max_depth	2–20	9
	min_samples_split	2–20	5
	min_samples_leaf	1–20	4.56
XGBoost	n_estimators	10–200	28.96
	max_depth	3–10	9
	learning_rate	0.01–0.3	0.1
	subsample	0.5–1.0	0.8
	colsample_bytree	0.5–1.0	0.56
CatBoost	iterations	10–200	135.20
	depth	3–10	9
	learning_rate	0.01–0.3	0.1
	l2_leaf_reg	1–10	9
LightGBM	n_estimators	10–200	89.23
	max_depth	3–10	7
	learning_rate	0.01–0.3	0.1
	subsample	0.5–1.0	0.86
	colsample_bytree	0.5–1.0	0.6

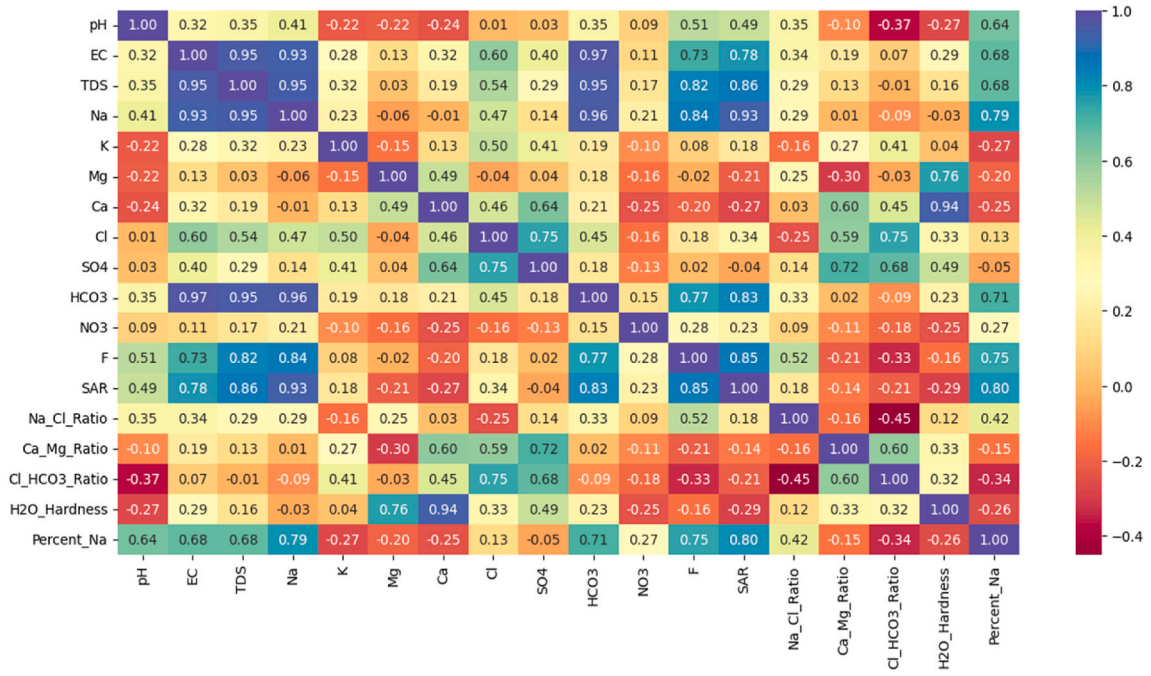


Fig. 2. Correlation matrix of the water quality features.

discouraging overly flexible posterior distributions that cannot be generalized to unseen data. KL divergence scaling factors were applied to balance adherence to the prior distributions with adaptability to the observed data. By estimating the posterior predictive distribution, the BDL captures both parameter uncertainty (epistemic) and inherent variability in the data (aleatoric) (Gal and Ghahramani, 2016; Hüllermeier and Waegeman, 2021; Kendall and Gal, 2017). All weights and biases were parameterized as probability distributions rather than fixed-point estimates, allowing the model to explicitly represent uncertainties in its predictions. For each input feature, predictive distributions were generated by performing stochastic forward passes and sampling weights from the learned variational posteriors, yielding mean class probabilities and variances (Kendall and Gal, 2017). The total predictive uncertainty is expressed as the sum of the aleatoric and epistemic contributions, which supports risk assessment and adaptive decision-making. Subsequently, the SHAP values were computed to improve the transparency and quantify the marginal contribution of each input feature to the predictive distribution (Lundberg and Lee, 2017). This probabilistic framework provides a transparent, reproducible, and theoretically grounded approach for WQI classification. The Bayesian paradigm offers a rigorous approach to quantifying model uncertainty and provides a framework for several regularization techniques and learning strategies (Josipin et al., 2022).

In the Bayesian interpretation, θ represents the parameters (weights and biases) of the neural network, and x and y denote the observed input–output data used to update prior beliefs about these parameters. The prior distribution $p(\theta)$ encodes initial assumptions regarding model complexity before observing data. Given the data, uncertainty is updated through the posterior distribution $p(\theta|x, y)$, which represents the distribution over network parameters conditioned on the observations. The likelihood function $p(y|x, \theta)$ describes the probability of the observed WQI class labels given the input physicochemical predictors and a particular realization of the network parameters. This likelihood captures aleatoric uncertainty arising from inherent variability and measurement noise in the data. The marginal likelihood $p(y|x)$, obtained by integrating over all possible parameter values, serves as a normalization constant and is generally intractable for deep neural networks. The Bayesian predictive distribution for WQI classification is obtained by marginalizing over the posterior distribution of the network param-

eters Eq. (17) and its integral is approximated using MC sampling from the posterior distribution (Eq. (18)):

$$p(x|y) = \int p(y|x, \theta) p(\theta|x, y) d\theta \quad (17)$$

$$p(y|x) \approx \frac{1}{N} \sum_{i=1}^N p(y|x, \theta_i), \theta_i \sim p(\theta|x, y) \quad (18)$$

where X denotes the vector of standardized physicochemical water-quality predictors, Y represents the WQI class labels, and θ_i are samples of the neural network parameters drawn from the posterior distribution. This formulation enables the Bayesian deep neural network (BDNN) to explicitly propagate parameter uncertainty into predictive uncertainty. As the posterior distribution $p(\theta|x, y)$ is analytically intractable for deep neural networks, variational inference was employed to approximate it using a tractable variational distribution $q(\theta|\phi)$, parameterized by variational parameters ϕ . These parameters were optimised by maximizing the Evidence Lower Bound (ELBO), defined as (Eq. (19)).

$$L_{ELBO}(\phi) = E_{\{q(\theta|\phi)\}} [\log p(y|x, \theta)] - KL(q(\theta|\phi) \| p(\theta)) \quad (19)$$

where $p(y|x, \theta)$ is the likelihood function and $KL(\cdot \| \cdot)$ denotes the Kullback–Leibler divergence between the variational posterior and the prior distribution $p(\theta)$.

Maximizing the ELBO establishes a principled trade-off between data fidelity and model complexity. The expected log-likelihood term promotes accurate data fitting, while the KL divergence term acts as a regularization penalty that discourages overly complex posterior distributions unless strongly supported by the data. This balance mitigates overfitting while enabling uncertainty-aware learning. This weakly informative prior regularizes model complexity while remaining sufficiently flexible to adapt to observed hydrochemical patterns. Model training was conducted using the Adam optimisation algorithm with a fixed learning rate. The network was trained for a predefined number of epochs using mini-batch gradient descent, and convergence was monitored through stabilization of the ELBO objective. Key training hyperparameters, including learning rate, batch size, number of hidden units,

and number of training epochs, were selected through preliminary testing to balance predictive performance and computational efficiency. The overall workflow of this work uses key environments of Python 3.11 with NumPy 1.25, TensorFlow 2.18.0, TensorFlow Probability 0.25.0, and SciPy 2.0.0.

3. Results and discussion

3.1. Geochemical patterns

Identifying the relationships among hydrochemical parameters is crucial for elucidating the key geochemical processes that control water quality. Correlation patterns were evaluated using statistical performance criteria, allowing a more precise characterization of interdependencies among geochemical parameters and their derived indices. Physicochemical parameters, evaluated against WHO guidelines, showed broad compliance with established drinking-water standards. Groundwater exhibited a mildly alkaline character, with conductivity and dissolved-solid levels reflecting mineralization shaped by infiltration pathways and residence time. Major cations showed consistent geochemical behavior, with calcium and magnesium remaining within permissible limits, potassium indicating agricultural inputs, and sodium showing localized enrichment. Anion concentrations were generally low, with chloride, nitrate, and sulfate signaling minimal contamination. Fluoride and bicarbonate reflected natural hydrochemical evolution and likewise remained within acceptable ranges. Correlation analysis provides a quantitative means to discern these interdependencies, revealing both direct and indirect linkages among ions and derived indices. The correlation heatmap (Fig. 2), provides a detailed analysis of the interdependencies between the measured physicochemical parameters and the derived variables. The Sodium Adsorption Ratio (SAR) exhibited strong positive correlations with sodium (Na^+ ; $r = 0.86$), electrical conductivity (EC; $r = 0.79$), and total dissolved solids (TDS; $r = 0.83$), confirming that Na^+ plays a central role in driving salinity-induced sodicity. These relationships align with established hydrogeochemical frameworks, wherein Na^+ enrichment commonly results from the dissolution of soluble minerals and anthropogenic contributions such as irrigation return flows or wastewater infiltration. EC and TDS are integrative indicators of ionic strength, and their co-variation with SAR confirms that overall salinity and sodicity hazards are inherently coupled, posing threats to soil structure, permeability, and agricultural sustainability. Calcium (Ca^{2+}) and magnesium (Mg^{2+}) showed moderate positive correlations ($r = 0.60$), but negligible direct associations with SAR (Ca^{2+} : $r = -0.04$; Mg^{2+} : $r = 0.07$) (Fig. 2). This observation reflects their dual role in contributing to the overall ionic content while simultaneously buffering sodicity hazards through mechanisms such as cation exchange and mineral precipitation equilibria. The derived ionic ratios further elucidate these diagnostic geochemical signatures. The Na^+/Cl^- ratio exhibited a modest positive correlation with SAR ($r = 0.18$), indicating potential ion-exchange processes or influences from freshwater dilution, whereas the negative correlation of SAR with the $\text{Cl}^-/\text{HCO}_3^-$ ratio ($r = -0.29$) suggests distinctive salinization mechanisms characterised by chloride dominance relative to bicarbonate (Fig. 2). Importantly, the percentage of sodium showed a strong positive correlation ($r = 0.79$), reinforcing its reliability as a composite indicator of sodicity hazards and its suitability for assessing irrigation (Fig. 2). Conversely, parameters such as pH and NO_3^- exhibited weaker correlations, highlighting the significance of flexible modeling techniques that can appropriately downweigh features with minimal predictive relevance. The observed correlation patterns indicate that predictive models trained on this dataset are likely to generalise well to new samples with similar geochemical properties. However, predictive efficacy may diminish when models are applied in hydrogeologically distinct regions or areas influenced by anthropogenic factors. Thus, the scientifically coherent geochemical patterns identified here enhance both the interpretability and operational value of

machine-learning predictions, facilitating transparent decision-making in water-quality monitoring, risk assessment, and targeted mitigation for sustainable agriculture. Building on the identified geochemical patterns, the following section evaluates the predictive performance of machine learning models for WQI estimation.

3.2. Machine learning prediction performance

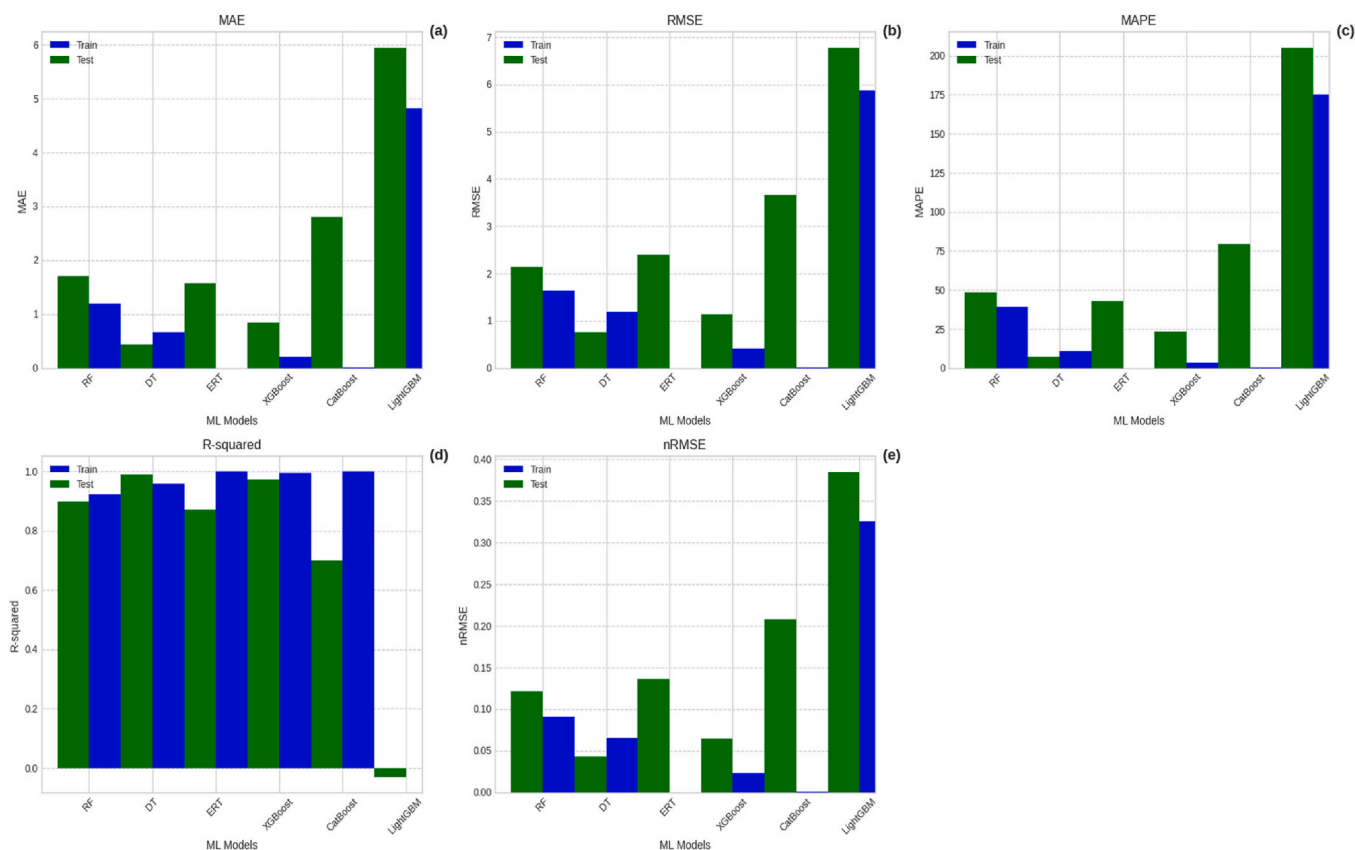
All ensemble machine learning models, such as DT, RF, ERT, XGBoost, CatBoost, and LightGBM were trained to predict the target variable with hyperparameters optimised using Bayesian optimisation (Table 1) to ensure a fair comparison and robust generalisation (Table 2). Bold values in Table 2 indicate best performance across models based on parameter sets (Table 1). Figs. 3 and 4 illustrate how the predictive accuracy and error patterns varied noticeably among the models. Fig. 3a–e shows the model error metrics MAE, RMSE, MAPE, nRMSE and R^2 for both the training and test sets. While ERT and CatBoost fit the training data almost perfectly ($R^2 \sim 1.0$), their test errors increased substantially, reflecting overfitting (ERT test $R^2 \sim 0.87$; CatBoost ~ 0.70). In contrast, DT and XGBoost maintained a high consistency between the training and test performances, indicating strong generalisation. The DT achieved the lowest test RMSE (0.747) and the highest R^2 (0.987), with prediction points tightly clustered along the 1:1 diagonal (Fig. 3a), demonstrating that simpler models can effectively capture dominant geochemical relationships, particularly when the predictor space is governed by a few strong variables. XGBoost also delivered excellent performance (RMSE = 1.134, $R^2 = 0.971$; Fig. 4b), highlighting the benefits of gradient boosting when rigorously tuned. In contrast, RF and ERT exhibited only moderate predictive power (RMSE ~ 2.1 – 2.4 , $R^2 \sim 0.90$ – 0.87 ; Fig. 3c and d), with broader scatter reflecting reduced precision. CatBoost and LightGBM performed substantially worse: CatBoost showed a higher prediction error (RMSE = 3.668, $R^2 = 0.698$; Fig. 4e), whereas LightGBM failed to generalise effectively, with a negative R^2 (-0.031) and RMSE exceeding 6.7 (Fig. 4f). These statistical indicators collectively quantify model accuracy (RMSE, MAE), systematic bias (MAPE), and explanatory power (R^2) across differing magnitudes and scales. All metrics were computed on an independent test set to provide an unbiased estimate of generalization performance. To diagnose overfitting, the study compared training and test errors, and models showing large discrepancies i.e. much lower training error relative to testing error were subject to further scrutiny or regularization.

The performance metrics summarised in Table 2 and Fig. 4 further confirm that increasing algorithmic complexity does not necessarily improve predictive performance; rather, it often leads to overfitting or instability, particularly in smaller environmental datasets. The superior accuracy and interpretability of DT show that transparent models can outperform more complex ensembles, providing reliable and explainable predictions essential for regulatory actions. Hence, the combination of Bayesian optimisation and thorough validation enabled all models to avoid severe underfitting or overfitting despite data constraints. The residual errors and R^2 in Fig. 4 support practitioners in diagnosing when and why models underperform, which is a critical step for credible risk assessment and adaptive monitoring. These results show that a higher model complexity does not guarantee better predictions. The optimised DT, a simple and transparent model for water quality management, outperformed more advanced ensembles by effectively capturing prediction variability. The superior performance of DT can be attributed to the dominance of a few strong geochemical drivers (Na^+ , EC, TDS, HCO_3^-), which align well with the threshold-based splitting mechanism of DT. More complex boosting methods tended to overfit on secondary predictors, especially given the modest dataset size, leading to poorer generalization. The superior performance of simple DT demonstrates that accurate and interpretable predictions do not always require computationally intensive models. The selected model demonstrated consistently strong predictive performance across multiple evaluation

Table 2

Comparative performance of Bayesian optimised machine learning models showing training and testing accuracies across multiple metrics.

Model	Train					Test				
	MAE	RMSE	MAPE	R ²	nRMSE	MAE	RMSE	MAPE	R ²	nRMSE
RF	1.192	1.643	39.121	0.922	0.091	1.712	2.133	48.723	0.898	0.121
DT	0.657	1.181	10.561	0.960	0.065	0.428	0.747	7.155	0.987	0.042
ERT	0.000	0.000	0.000	1.000	0.000	1.565	2.403	42.931	0.870	0.136
XGBoost	0.206	0.415	3.235	0.995	0.023	0.846	1.134	23.558	0.971	0.064
CatBoost	0.011	0.013	0.426	1.000	0.001	2.800	3.668	79.479	0.698	0.208
LightGBM	4.815	5.869	175.186	0.000	0.325	5.943	6.778	205.238	-0.031	0.384

**Fig. 3.** Machine Learning Model Performance Metrics bar plots on Train and Test sets.

metrics, confirming its ability to capture dominant hydrochemical controls on WQI. This efficiency reduces energy demand, supporting sustainable AI practices and enabling continuous monitoring in resource limited regions. Hence, a reliable and interpretable DT can guide targeted interventions, balancing predictive power and explainability to integrate ML into sustainable monitoring, and build confidence in AI-based solutions followed by XGBoost.

Overall, this evidence supports the adoption of hybrid workflows that balance accuracy and interpretability, ensuring that AI-powered water quality assessments are scientifically rigorous, actionable, and aligned with sustainable management goals. Bayesian optimisation improved the predictive accuracy and reduced the overfitting risk, as evidenced by the lower RMSE and consistent R² across the folds (Table 2). Its efficiency makes it suitable for environmental applications, in which computational resources may be limited. This evaluation strategy reveals that generalizability across heterogeneous hydrogeological settings, model parameters and structures. By combining ML with robust optimisation and transparent evaluation, the shift from reactive estimation to proactive decision support can be achieved, empowering more informed strategies and enhancing the resilience of

water resource management. The ML-based predictions capture nonlinear interactions among hydrogeochemical parameters that are not explicitly represented in the arithmetic WQI formulation, resulting in improved predictive accuracy and interpretability. The superior performance of DT relative to more complex ensemble methods can be attributed to the structured nature of the dataset and the dominance of a limited number of key physicochemical drivers. In such settings, simpler models can achieve strong generalization by avoiding unnecessary complexity and overfitting, highlighting that model interpretability and data characteristics may be as important as algorithmic complexity.

While these models demonstrated impressive predictive performance, particularly the DT outputs, remain point estimates that do not explicitly quantify predictive uncertainty. This limitation is important because aquifer systems are inherently dynamic, with measurement noise, temporal variability, and incomplete sampling, introducing uncertainty that deterministic predictions alone cannot capture. Consequently, integrating probabilistic learning improves confidence and reliability in water quality assessments. This capability enables predictions to be accompanied by explicit confidence intervals and probabilistic thresholds, thereby supporting risk-aware management

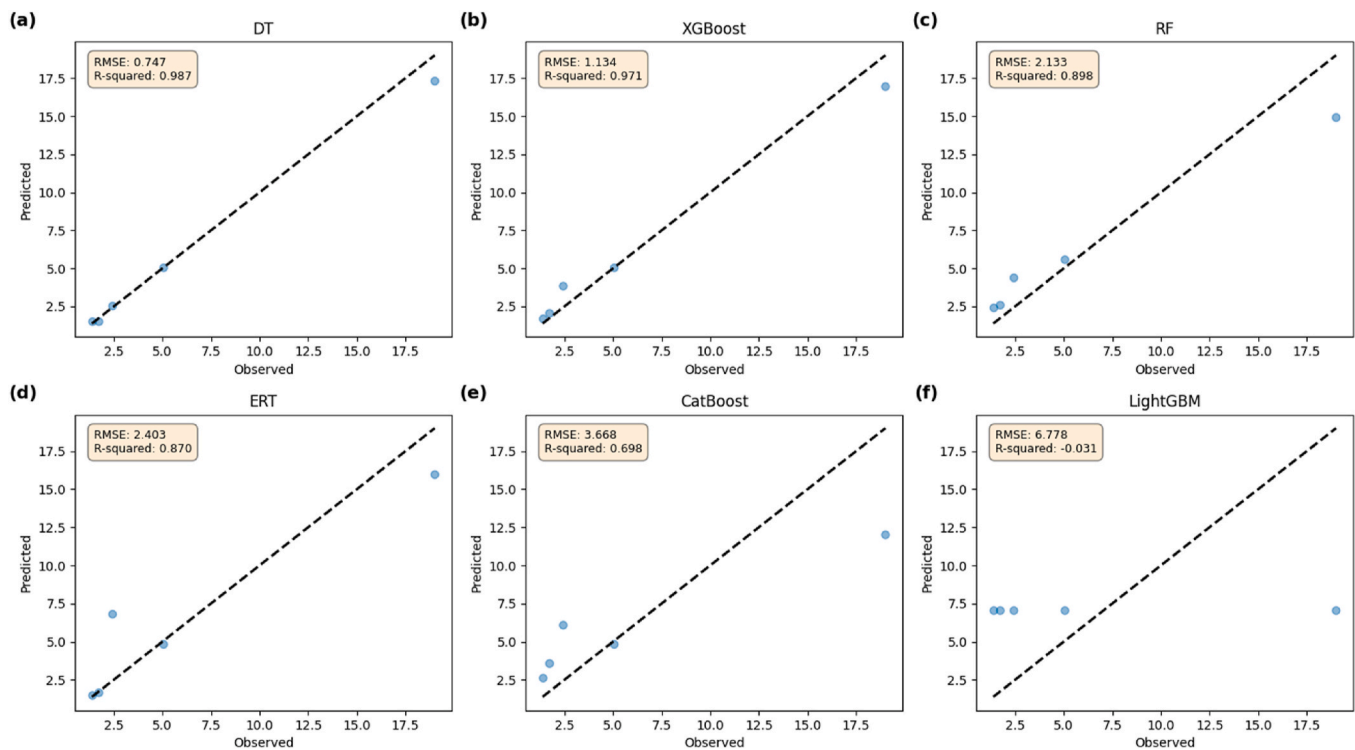


Fig. 4. Machine Learning performance plots of the test set.

decisions. Incorporating Bayesian inference thus strengthens transparency, addressing stakeholder concerns regarding overconfidence and supporting adaptive monitoring frameworks that align with SDGs for sustainable water resource management and public health protection.

3.3. Model optimisation and loss convergence

The BDL model demonstrated robust learning dynamics and high-accuracy predictive performance. The cross-entropy loss curve declined smoothly over 1000 training epochs, with the training and validation losses remaining closely aligned, indicating a strong generalisation without overfitting (Gal and Ghahramani, 2016). The steep

initial drop in (Fig. 5) reflects the rapid learning of dominant physico-chemical patterns, whereas slower decline in the later epochs captures subtler relationships. At initialization, the losses exceeded 40 because of random parameter weights. The losses plateaued at approximately epoch 400, suggesting that the residual error primarily reflected the aleatoric uncertainty inherent in the data rather than model deficiencies. A low final cross-entropy loss confirms convergence and produces well calibrated probabilistic outputs (Blundell et al., 2015). This stability is essential in Bayesian frameworks, where uncertainty quantification depends on accurate probability estimates and effective posterior approximations through variations. Although minimal improvements occurred after epoch 600, an earlier stop could have

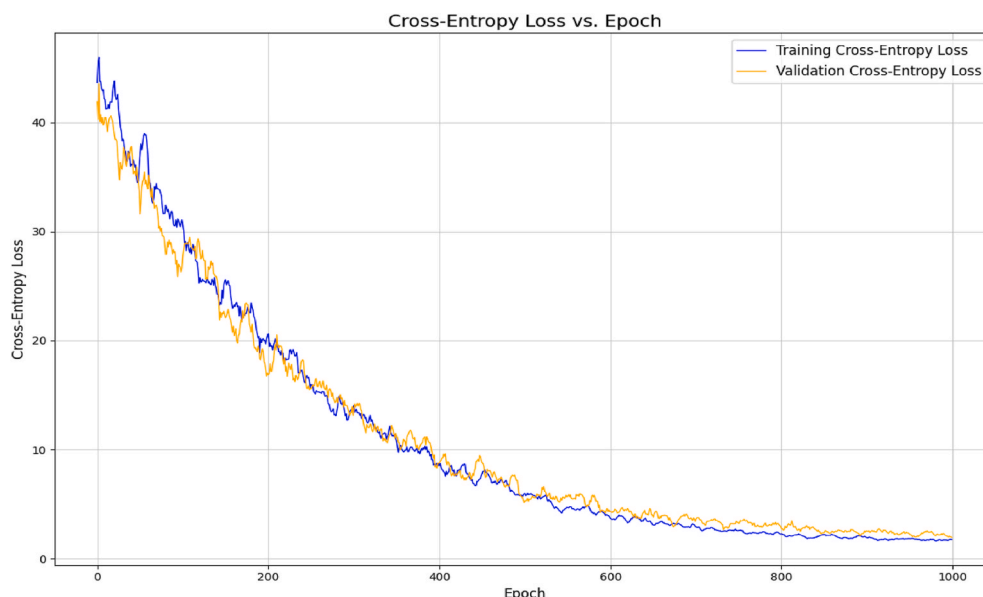


Fig. 5. Predictive cross-entropy quantifying uncertainty in the model's probabilistic WQI predictions.

achieved a similar performance with a lower computational cost (Fig. 5). This clear pattern (Fig. 5) of feature contributions reinforces the confidence in the model interpretability and alignment with the domain understanding.

This plot tracks the cross-entropy loss during training, illustrating how predictive uncertainty decreases as the model learns. In probabilistic WQI modeling, lower cross-entropy corresponds to sharper and more confident predictive distributions. High loss in early epochs reflects broad, high-entropy outputs, whereas the sustained decline in both the training and validation curves indicates that the model is progressively forming more concentrated and informative probability estimates. The close agreement between the two curves shows that this reduction in uncertainty generalizes to unseen data, avoiding overconfidence on the training set and signaling stable probabilistic behavior across the feature space. As the model internalizes underlying environmental patterns, its predictive distributions become increasingly well calibrated. This behavior is especially important for WQI applications, where variables exhibit inherent stochastic variability. The converging cross-entropy curves confirm that the model captures the dominant structure in the data and produces reliable probabilistic outputs suitable for operational decision support. Overall, the plot shows a steady decline in predictive entropy, indicating that the model's probabilistic WQI estimates become more confident as training progresses. The smooth convergence, and interpretable attributions together demonstrate that the framework effectively captures underlying relationships, enabling reliable, uncertainty-aware predictions essential for water-quality assessment and regulatory decision-making.

3.4. Model prediction interpretability

SHAP (SHapley Additive exPlanation) analysis provided evidence of consistent learning and meaningful feature prioritisation. Fig. 6 presents a SHAP summary bar chart ranking the importance of each physico-chemical variable in predicting WQI classes using BDL model. Each bar shows the mean absolute SHAP value across the test samples, indicating

the average contribution of each feature to the model predicted probabilities. The summary plot (Fig. 6) shows that Cl^- , HCO_3^- , Na^+/Cl^- ratio, K^+ , and TDS are the most influential predictors, which is consistent with the established hydrochemical processes driving salinity- and alkalinity-related water quality degradation. Higher HCO_3^- and Na^+/Cl^- ratios contributed to the prediction of poorer WQI classes, whereas lower EC and TDS were linked to higher-quality water categories. HCO_3^- reflects its role in buffering alkalinity and indicates geochemical processes and anthropogenic inputs such as agricultural runoff and carbonate dissolution (Qadir et al., 2006). High bicarbonate levels helped distinguish higher WQI classes from lower WQI. The Na^+/Cl^- ratio is also important for capturing salinity dynamics linked to evaporation, rock-water interactions, and contamination from irrigation return flows and wastewater discharge. Other top-ranked features included K^+ , Cl^- , TDS, F^- , Na^+ , and EC, underscoring salinity as a central driver of water quality degradation (Scanlon et al., 2017).

Feature contributions varied across classes. For instance, bicarbonate and Na^+/Cl^- ratio strongly influenced the predictions of intermediate quality classes 2 and 3, whereas EC and sodium contributed more to the identification of poor-quality classes 0 and 1. Mid-ranking predictors (Fig. 6), such as Mg^{2+} , SO_4^{2-} , SAR, hardness, and percent Na^+ , were aligned with the known hydrochemical processes affecting the potability and irrigation suitability. Lower-ranking variables (Fig. 6), such as pH, NO_3^- , Ca^{2+} , and $\text{Cl}^-/\text{HCO}_3^-$ ratio, contributed modestly but retained their predictive value. SHAP analysis showed that HCO_3^- , Na^+/Cl^- ratio, EC, and TDS had the strongest influence on the predictions. High influence features (HCO_3^- , Na^+/Cl^- , K^+ , Cl^- , TDS), and their dominant role is high relative to low influence (pH, NO_3^- , Ca^{2+} , $\text{Cl}^-/\text{HCO}_3^-$) predictors. Samples with intermediate values for these features were more difficult to classify confidently, resulting in higher uncertainties. SHAP interpretability validates the scientific relevance of the predictors and highlights the opportunities for targeted monitoring of bicarbonate, salinity indicators, and conductivity to detect early deterioration. This level of transparency ensures that predictions are not only statistically robust but also scientifically credible, which is an essential requirement

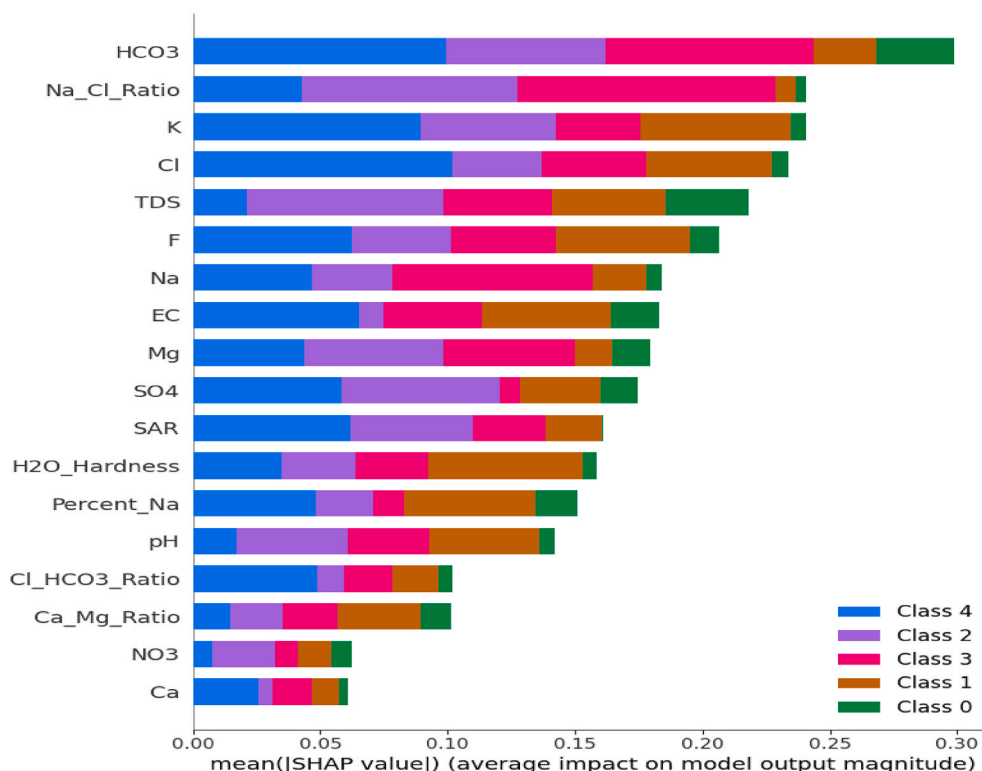


Fig. 6. Mean SHAP value summary plot showing the overall feature importance of the WQI.

for regulatory acceptance and gaining stakeholder trust (Lundberg and Lee, 2017).

The SHAP analysis for the tuned Decision Tree model provides a clear view of the mechanistic drivers of SAR in the dataset. Features with large SHAP magnitudes are the primary contributors to SAR variability. High-value points that push SHAP values to the right indicate strong positive contributions to SAR, which is common for salinity- and ion-concentration metrics. Low-value points that push SHAP values left reflect negative contributions or SAR-suppressing effects. These predictors should be prioritized for high-precision monitoring, routine sensor calibration, and targeted process-control interventions. Characterizing threshold behavior and nonlinear responses within these features can support proactive water-treatment strategies. The tuned Decision Tree Regressor emerged as the strongest operational candidate, exhibiting a generalization gap of ΔR^2 (Train-Test) = -0.028 and an RMSE ratio (Test/Train) of 0.633 evidence of strong predictive reliability without pathological overfitting. The modest train-test divergence, combined with consistently low residuals, indicates a stable decision structure that generalizes well across the feature domain. Given its accuracy, interpretability, and generalization pattern, this model is well-suited for real-time SAR prediction pipelines and operational decision support. Targeted statistical and chemical analyses can help elucidate ion-ion interactions affecting adsorption dynamics, potential nonlinearities revealed by SHAP distributions, and coupling effects among key features. These findings support model refinement, water-treatment optimisation, and the development of expert rules to complement the ML pipeline. The SHAP analysis further highlights the dominant chemical or related drivers of SAR, clarifying how high or low predictor values exert strong reversing influence on predictions. These insights interpret right into operational guidance: prioritize accurate monitoring of the most influential predictors, develop understanding of their physicochemical relationships to SAR, and maintain strict data

quality controls. For long-term reliability, implement continuous performance tracking, periodic retraining, and drift detection as water conditions change. Augmenting the feature set with environmental or geochemical data and exploring ensemble approaches may further enhance robustness. Integrating the model into water management operations enables proactive SAR control, early warning for high-risk conditions, and data-driven optimisation of treatment strategies.

3.5. Probabilistic prediction uncertainty

3.5.1. Uncertainty across samples

In water quality assessment, quantifying uncertainty is key to determining the reliability of the input features (Uddin et al., 2023a). The mean predicted probability across multiple stochastic forward passes of the BDL model. The relatively higher bars (Fig. 7), where the predicted probabilities match the true WQI classes, indicate that the model is generally confident. This pattern aligns with the cross-entropy loss curve, which decreases smoothly over the training epochs without divergence between the training and validation losses. The low final cross-entropy loss suggests that the model successfully learned to assign high probability predictions to the correct labels. Fig. 7 shows the predicted class probabilities for each WQI category, along with the epistemic uncertainty depicted as standard deviation error bars derived from the Monte Carlo (MC) posterior sampling. MC dropout and variational inference was implemented by sampling the posterior 50 times at prediction stage to estimate epistemic uncertainty. Epistemic uncertainty, arising from model parameter variability, was quantified using posterior variance across 50 MC samples, while aleatoric uncertainty, reflecting data noise, was estimated from predictive variance within class probability distributions. Practically, wide epistemic intervals indicate insufficient training data in certain regions of feature space, guiding regulators to prioritize additional sampling. In contrast, wide

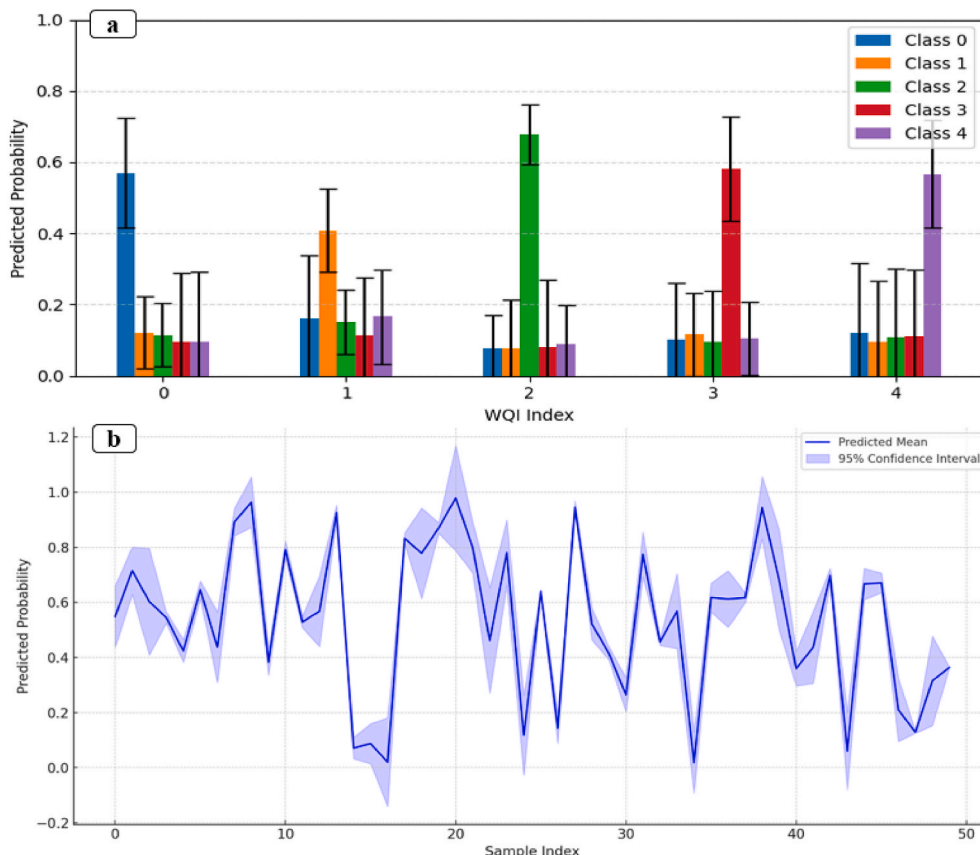


Fig. 7. (a). Predicted Class Probabilities with Epistemic Uncertainty; (b). Prediction plot with a 95 % confidence interval of Uncertainty Across Samples.

aleatoric intervals capture inherent variability in water chemistry that cannot be reduced by more data, highlighting the need for risk thresholds with safety margins rather than expectations of certainty. This distinction is critical for designing monitoring strategies that balance data collection with realistic management of environmental variability. In (Fig. 7a), a clear separation between the dominant predicted class and the lower probabilities of the others is evident, especially for WQI Classes 0 and 2, where the mean probabilities exceeded 0.5 and the uncertainty remained moderate. The wider error bars for Classes 1 and 3 indicate less confident predictions, reflecting the limited data coverage and overlapping feature distributions (Fig. 7a). This corresponds to the mid-training phase, when the loss reduction slows down as the model encounters more ambiguous samples. The low cross-entropy loss and strong class separation insights demonstrate that BDL effectively prioritizes relevant predictors while transparently conveying uncertainty. Practically, this supports risk-aware decision-making: predictions with high uncertainty can prompt additional sampling, whereas confident outputs can inform operational actions. Fig. 7 illustrates how the BDL achieves an accurate, interpretable, and uncertainty-aware classification of water quality status.

The synergy between high-probability separation and quantified uncertainty underscores the advantages of Bayesian modeling for adaptive groundwater management. Fig. 7b illustrates the predictive performance of the BDL model applied to probabilistic classification, emphasising the uncertainty quantification across the test samples. The central blue line shows the mean predicted probability, which was computed by aggregating multiple MC draws from the posterior distribution of the model weights. This represents the best estimate of the probability that each observation belongs to a given class. The surrounding shaded area depicts the 95 % confidence interval (Fig. 7b), which quantifies the epistemic uncertainty arising from the limited data and uncertainty in the learned parameters. This uncertainty was estimated as the variability in the predictions across stochastic forward passes using variational inference. Narrow confidence intervals (Fig. 7b) indicate high certainty and stability in the predictions, whereas wider intervals indicate substantial uncertainty and potential inconsistency. Conversely, tightly bound intervals reflect high confidence and reproducibility. Predictions near the critical thresholds with wide intervals indicate that further sampling is necessary. Moreover, Fig. 7b offers diagnostic insights into the model behavior by showing how uncertainty varies across samples, thereby helping practitioners identify where additional data could improve confidence.

Uncertainty-aware WQI predictions provide actionable guidance for water regulators by highlighting where and how monitoring resources should be deployed. High epistemic uncertainty indicates gaps in training data, enabling agencies to prioritize additional sampling, optimize monitoring frequency, and justify the installation of new observation wells in underrepresented regions. In contrast, high aleatoric uncertainty reflects intrinsic variability in water chemistry, such as at wastewater discharge sites, where precautionary thresholds and safety margins should be adopted rather than expecting precise predictions. By distinguishing between these two uncertainty sources, regulators can allocate remediation resources more efficiently and design adaptive monitoring strategies that balance cost, risk, and environmental protection. The findings highlight that incorporating uncertainty into WQI predictions provides a practical basis for adaptive groundwater monitoring and management. By quantifying predictive uncertainty, the proposed framework enables decision-makers to identify regions of high risk or data scarcity, prioritize additional sampling in zones with high uncertainty, and allocate monitoring resources more efficiently. These insights support risk-based management and can guide the formulation of dynamic water-quality surveillance programs that respond to evolving hydrogeochemical conditions. Such uncertainty-aware approaches represent a move from static assessments toward evidence-based and resilient groundwater governance.

3.5.2. Uncertainty across WQI classes

The figure (Fig. 8) illustrates the distribution of total predictive uncertainty produced by the Bayesian deep learning model across the WQI classes in the test set. A clear class-dependent structure in uncertainty is observed, indicating that predictive confidence is not uniform across water quality categories. Lower and intermediate WQI classes (Classes 2 and 3) exhibit the lowest uncertainty, suggesting that these conditions are well represented in the training data and correspond to more stable and homogeneous hydrogeochemical regimes. In contrast, poorer and extreme water quality classes (particularly Class 4) show substantially higher uncertainty, reflecting increased epistemic uncertainty associated with data scarcity, higher variability, or more complex hydrochemical processes in these regimes. Moderate uncertainty levels in Classes 0 and 1 further indicate transitional conditions where class boundaries may overlap, leading to reduced model confidence. This pattern demonstrates that the Bayesian framework effectively captures heterogeneity in model confidence across WQI states, rather than producing uniformly confident predictions. Importantly, higher uncertainty in degraded water quality classes highlights areas where predictions should be interpreted with caution and where additional monitoring or data collection may be most valuable. Overall, the results confirm that uncertainty estimates are physically meaningful and provide actionable insights for risk-aware water quality assessment and adaptive management. This reinforces the Bayesian perspective that predictions are distributions, not single values, providing decision-makers with transparent and interpretable information to support risk-aware resource-management. This capability highlights a key advantage of Bayesian approaches in environmental modeling: the ability to propagate and explicitly quantify predictive uncertainty, rather than relying on point estimates. In WQI prediction, this is crucial because assessments inform regulatory and public health decisions regarding water quality. Epistemic uncertainty, which dominated in underrepresented feature regions, indicates where regulators should prioritize additional sampling (in poorly characterized aquifer zones). Aleatoric uncertainty, reflecting intrinsic variability in water chemistry, signals that even extensive sampling cannot fully eliminate uncertainty, and thus risk thresholds should incorporate these confidence intervals. By quantifying epistemic and aleatoric uncertainty, the framework equips regulators and water managers with confidence intervals that support risk-aware decisions, guide additional sampling where model uncertainty is high, and reduce misclassification risks in compliance settings.

3.5.3. Bias diagnostics and uncertainty dispersion

For water quality assessment, reliable predictions must be accompanied by transparent uncertainty estimates that distinguish between systematic model error and inherent environmental variability. Beyond class-wise uncertainty patterns, the reliability of the proposed Bayesian framework is further supported by an explicit analysis of prediction bias and uncertainty dispersion (Fig. 9). The distributions of the bias mean (location parameter) (Fig. 9a), indicate that model predictions are centered close to zero bias across the test dataset, demonstrating the absence of systematic over- or underestimation of WQI classes. This behavior suggests that the learning process effectively captures dominant hydrochemical relationships without introducing persistent directional error, a critical requirement for robust environmental prediction. The distribution of the bias standard deviation (scale parameter) (Fig. 9b), exhibits a narrow and well-constrained range, indicating stable uncertainty estimates across samples. Such consistency reflects a well-calibrated probabilistic model in which predictive uncertainty responds coherently to variations in input space rather than fluctuating arbitrarily. Importantly, the absence of pronounced heavy tails or extreme dispersion in the scale parameter implies that uncertainty estimates are not inflated by noise or model instability, but instead reflect meaningful epistemic and aleatoric contributions. Collectively, the near-zero bias and stable uncertainty dispersion demonstrate that prediction errors are primarily driven by localized hydrogeochemical complexity

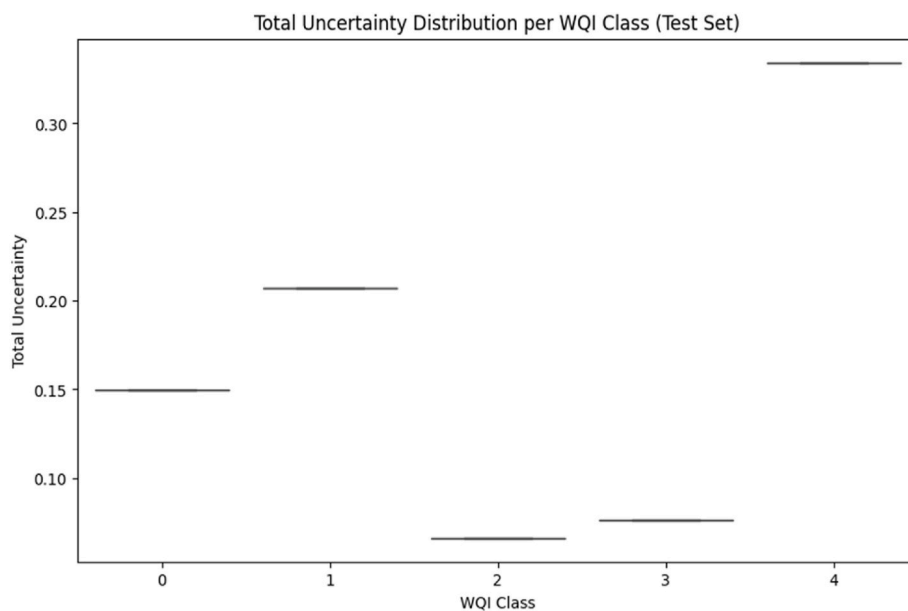


Fig. 8. Distribution of total uncertainty across WQI classes in test dataset.

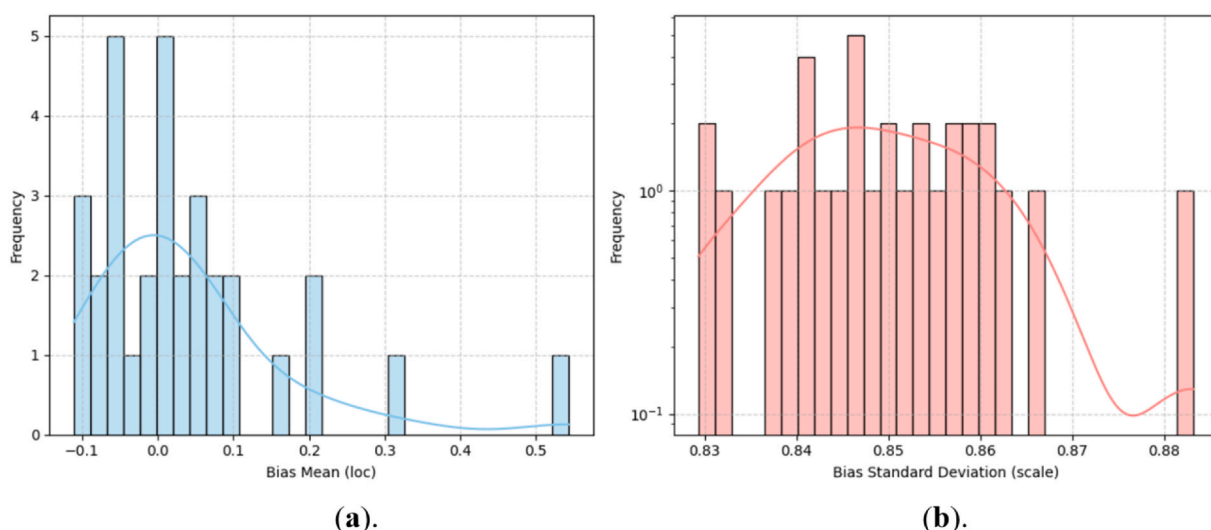


Fig. 9. Distributions of (a). mean, (b). standard deviation biases illustrating error behavior of predictions on the test set.

rather than structural model deficiencies. This dissimilarity is particularly important in heterogeneous groundwater systems, where uncertainty should arise from unaffected environmental variability rather than methodological artifacts. The near-zero bias mean and tightly concentrated bias variance are consistent with stable convergence, indicating that the variational posterior has achieved an appropriate balance between likelihood fit and regularization, resulting in unbiased and well-calibrated predictive uncertainty. Consequently, these diagnostics reinforce the credibility of the Bayesian framework for uncertainty-aware WQI assessment, providing confidence that both predictions and associated uncertainty estimates are scientifically suitable for risk-informed water quality management.

3.6. WQI category sensitivity and specificity

Fig. 10 presents the Receiver Operating Characteristic (ROC) curves for the predicted WQI classes. Each curve illustrates the trade-off between the true positive rate (sensitivity) and the false positive rate across different probability thresholds. The Area Under the Curve (AUC) values

quantified the discriminative performance of each class (Fig. 10). Classes 0 and 3 (Fig. 9) achieved the highest AUC scores (both 0.90), indicating excellent separability and demonstrating that the model could reliably distinguish samples with the most distinctive physicochemical properties. Classes 1 and 2 also showed strong performance, with AUCs of 0.88 and 0.86, respectively, confirming the model's ability to rank positive instances ahead of negatives across thresholds (Fig. 10). In contrast, Class 4 yielded the lowest AUC (0.77), suggesting that predictions for this category are more challenging because of the overlapping feature distributions with the intermediate classes. Nevertheless, all ROC curves lie well above the diagonal reference line, confirming a predictive power that is substantially better than that of random guessing. The steep initial ascent for classes 0, 1, and 3 (Fig. 10) indicated high sensitivity at low false-positive rates, supporting their use for early warning and screening. Fig. 10 highlights the value of combining probabilistic outputs with uncertainty metrics, particularly in marginal scenarios. The model demonstrated consistently strong predictive performance across multiple evaluation metrics, confirming its ability to capture dominant hydrochemical controls on WQI. Overall,

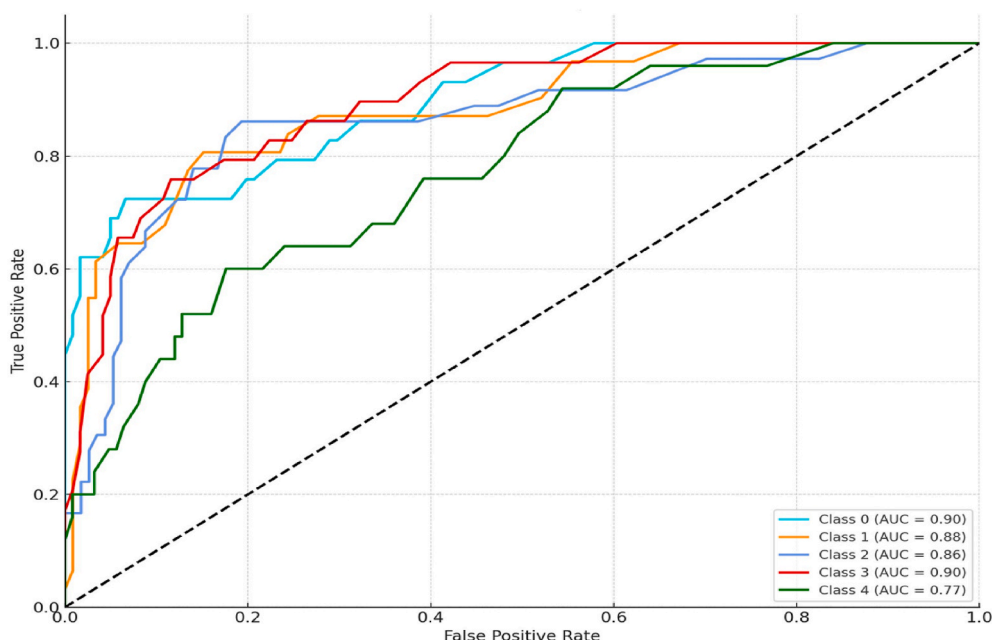


Fig. 10. Roc curves demonstrating the discriminative performance of models.

ROC analysis demonstrated that the Bayesian model effectively classified water quality categories and provided reliable, interpretable predictions for evidence-based water resource management.

3.7. Inferences for environmental sustainability

This study demonstrates that effective water quality prediction does not require complex and computationally intensive models. Large ensemble models typically require substantial training time, high memory, and more energy than simpler tree-based approaches (Prokhorenkova et al., 2018; Strubell et al., 2020). In contrast, the optimised DT trained rapidly, consumed minimal computational resources, and produced transparent decision rules without sacrificing performance, directly reducing the carbon footprint of training (Hasan et al., 2025), which is valuable for continuous monitoring in data limited settings. The transparent structure of the DT allows managers to trace predictions to explicit threshold values, building confidence and supporting more responsible decisions. While XGBoost achieved high accuracy, its limited interpretability and overfitting risks hinder practical deployment (Chen and Guestrin, 2016). The BDL framework enabled accurate WQI prediction with transparent uncertainty quantification, while the modest dataset size favored simpler models that effectively mitigated overfitting. Unlike deterministic models, BDL provides probabilistic outputs that convey confidence and variability, allowing decision-makers to prioritize sampling in uncertain wastewater treatment and avoid unnecessary interventions where predictions are reliable. This approach conserves monitoring resources and reduces computational overhead while maintaining high performance without the energy requirements of large ensembles. BDL illustrates how advanced machine learning can enhance sustainability by balancing precision and efficiency. These qualities advance water quality assessment and demonstrate how modern AI can improve efficiency and environmental sustainability while reducing the ecological footprint of digital technologies in line with sustainable development goals. The study uniquely integrates deterministic ensemble ML, XAI, and probabilistic BDL into a unified workflow for WQI prediction simultaneously addressing accuracy, interpretability, and uncertainty dimensions rarely combined in previous water quality studies. It is the first to demonstrate that a simple, interpretable DT can outperform more complex ensembles in WQI prediction, while coupling SHAP analysis with BDL to provide

both transparent feature attributions. Unlike fixed-weight WQI indices, ML models adaptively capture nonlinear feature interactions, improving predictive performance and flexibility.

Although this study demonstrates the potential of integrating ensemble ML, SHAP-based XAI, and BDL for WQI prediction, its generalization to other hydrogeological settings will require retraining and validation using diverse local datasets. The current analysis has limited geographic scope, sample size, and temporal coverage; while it effectively captures local geochemical variability, external validation across multiple basins and heterogeneous datasets is essential to enhance generalizability. The BDL framework offers valuable uncertainty quantification but entails considerable computational demands, which may restrict its application in resource-limited contexts. Furthermore, the dominance of a few key predictors improves interpretability yet may reduce robustness when inputs are noisy or incomplete. Future work should extend validation to broader hydrogeological settings, incorporate independent datasets, and develop computationally efficient implementations to enhance scalability. Despite these limitations, the proposed workflow is adaptable to data-scarce regions, supporting evidence-based water quality governance and directly contributing to Sustainable Development Goal 6 (Clean Water and Sanitation).

4. Conclusions and future perspectives

4.1. Conclusions

This study demonstrates that integrating ensemble machine learning with Bayesian deep learning enables accurate, interpretable, and uncertainty-aware prediction of the WQI. SHAP analysis identified the dominant physicochemical drivers of water quality, while probabilistic modeling delivered actionable estimates of predictive uncertainty. The results show that combining deterministic machine learning with probabilistic Bayesian deep learning provides a robust and adaptive framework for WQI assessment and uncertainty quantification. Among the evaluated models, the optimised decision tree achieved superior accuracy and interpretability compared with more complex ensemble methods when key geochemical controls were well characterized, highlighting that model simplicity when paired with rigorous hyperparameter optimisation can yield strong predictive performance. This finding demonstrates that accurate and interpretable WQI predictions

do not necessarily require computationally intensive models, thereby reducing energy demand, supporting sustainable AI practices, and enabling continuous monitoring in data-scarce regions. The Bayesian deep learning framework further enhanced the analysis by coupling high predictive accuracy with credible uncertainty quantification, facilitating transparent and risk-aware decision-making. Bayesian optimisation improved model generalization without incurring unnecessary computational costs, while SHAP-based interpretability clarified the influence of individual physicochemical parameters. By integrating explainable AI with Bayesian inference and variational posterior sampling, the proposed workflow captures multiple components of predictive uncertainty that are rarely addressed in water quality studies. Overall, this framework offers a scalable and uncertainty-aware pathway for strengthening environmental management, protecting public health, and enhancing adaptive capacity under increasing pressures on groundwater systems.

4.2. Limitations and future perspectives

Building on the findings and limitations identified, several strategic directions can guide future research to enhance the robustness, transferability, and operational value of machine-learning-based water quality assessment. The following recommendations and perspectives provide directions to strengthen the generalization capacity and practical relevance of the proposed framework. (i) Extend spatial and temporal validation: Future studies should validate the proposed framework across diverse hydrogeologic settings and climatic zones. This will help test model transferability and robustness to data heterogeneity, seasonal variability, and differing land use patterns. Long-term time-series integration can further reveal the temporal evolution of uncertainty and water quality trends. (ii) Integrate multi-source and real-time data streams: Incorporating high-resolution remote sensing products, and climate reanalysis datasets can improve spatiotemporal coverage and support near-real-time water quality prediction. Coupling these data with probabilistic data assimilation frameworks will enhance the capacity for adaptive water resource monitoring under changing environmental conditions. (iii) Advance physics-guided and hybrid learning frameworks: embed physical constraints or process-based priors within Bayesian deep learning architectures. Such physics-guided integration can constrain model extrapolation, reduce epistemic and aleatoric uncertainty, and strengthen physical interpretability. (iv) Expand to emerging and complex contaminants: The current framework can be extended to assess emerging pollutants where limited monitoring data aggravate uncertainty. Bayesian ensemble models can provide probabilistic risk maps and quantify the confidence bounds of contaminant predictions. Integrate remote sensing and real-time monitoring, and apply Bayesian ensembles to emerging contaminants such as microplastics and polycyclic aromatic hydrocarbons (PAHs). (v) Leverage sustainable and energy-efficient AI: Considering the environmental footprint of AI computation, future work should evaluate the trade-offs between model complexity, accuracy, and energy demand. Model refinement strategies can reduce computational costs while maintaining interpretability and uncertainty. (vi) Expanding the feature space with additional explanatory hydrological variables (flow rate), geochemical descriptors, spatial data (well depth, geology, aquifer characteristics), and weather or evapotranspiration metrics to strengthen physical interpretability and improve robustness under evolving environmental conditions. Overall, future research should extend validation across basins, while advancing toward integrative, physics-guided, and uncertainty-aware AI ecosystems that unify data from sensors, satellites, and models within transparent decision frameworks. Such convergence will support adaptive, evidence-based water management aligned with SDG 6 the broader vision of climate-resilient environmental governance.

CRedit authorship contribution statement

Tarekegn Dejen Mengistu: Writing – review & editing, Writing –

original draft, Visualization, Validation, Methodology, Investigation, Conceptualization, Formal analysis, Software. **Il-Moon Chung:** Writing – review & editing, Validation, Resources, Supervision, Funding acquisition. **Sun Woo Chang:** Writing – review & editing, Validation, Resources.

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Declaration of competing interest

The authors declare that they have no any competing interests.

Data availability

Data will be made available on request.

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